

The Bloom syndrome helicase (BLM) is catalytically inactive in Bloom syndrome (BS), a genetic disorder that leads to genomic instability characterized by excessive genetic recombination events in somatic cells. Research studies have indicated that BLM is a protein known as a replication stress response factor that aids in DNA synthesis repair of stalled replication forks during replication stress (arrest). To investigate this further, an experiment was conducted using cell lines derived from a patient with BS. Two groups of cells were generated, which initially differed only by BLM status. BLM⁺ cells were generated using cDNA cloning (ie, vector with BLM cDNA transfected). In contrast, BLM⁻ cells were transfected with an empty vector. Specific mutations, BLM-K695T and BLM-T99A, were also studied in stable transfected cells.

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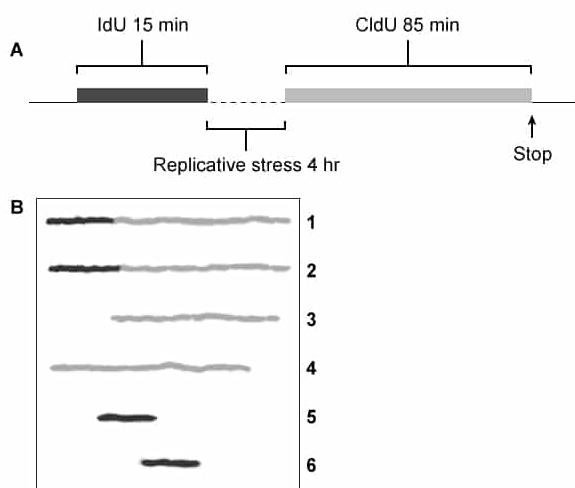


Figure 1 (A) Experimental protocol and (B) fluorescent imaging of a microscope slide showing incorporation of IdU (black lines) and CldU (gray lines) into DNA fibers isolated from a group of BLM⁺ and BLM⁻ cells, with single fibers depicted in each row

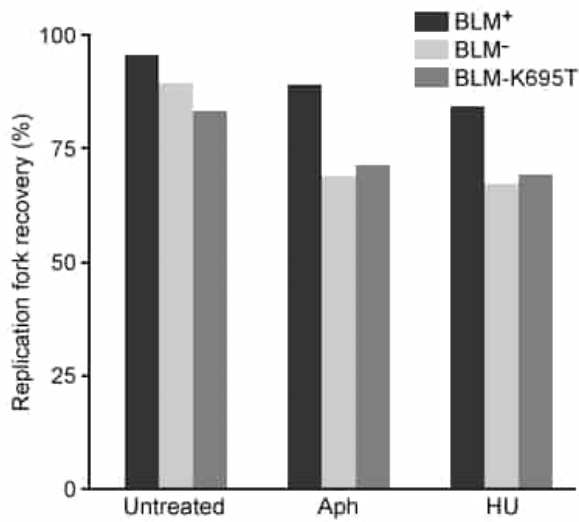


Figure 2 Recovery of stalled replication forks in cells expressing BLM⁺, BLM⁻, and the mutation BLM-K695T (antagonistic to wild-type)

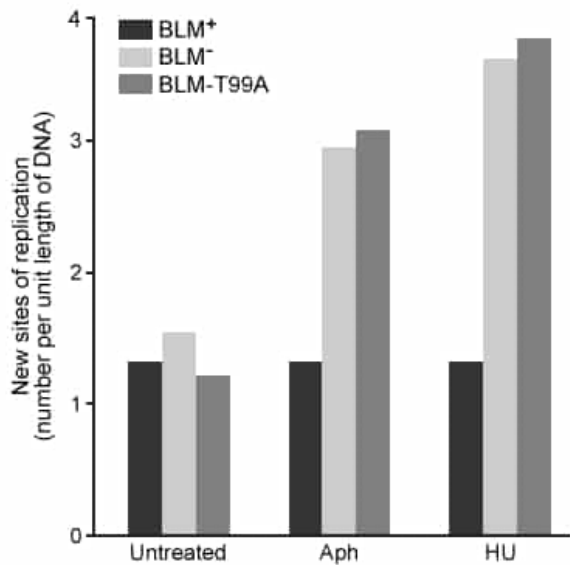


Figure 3 New sites of replication in cells expressing BLM⁺, BLM⁻, and the mutation BLM-T99A (T99 phosphorylated after replication arrest)

The BLM peptide sequence was used to search for similar peptides in various species using an amino acid sequence homology (similarity) database. BLM and other peptides were found to share similar amino acid sequences only in the DNA helicase domains. The results of sequence homology with BLM are shown in the table below.

Species	Peptide	Sequence homology	DNA helicase domains identified
<i>Saccharomyces cerevisiae</i> (yeast)	SGS1	43%	I, Ia, II, III, IV, V, VI
<i>Escherichia coli</i> (bacteria)	recQ	42%	I, Ia, II, III, IV, V, VI
<i>Homo sapiens</i> (human)	RECQL	43%	I, Ia, II, III, IV, V, VI

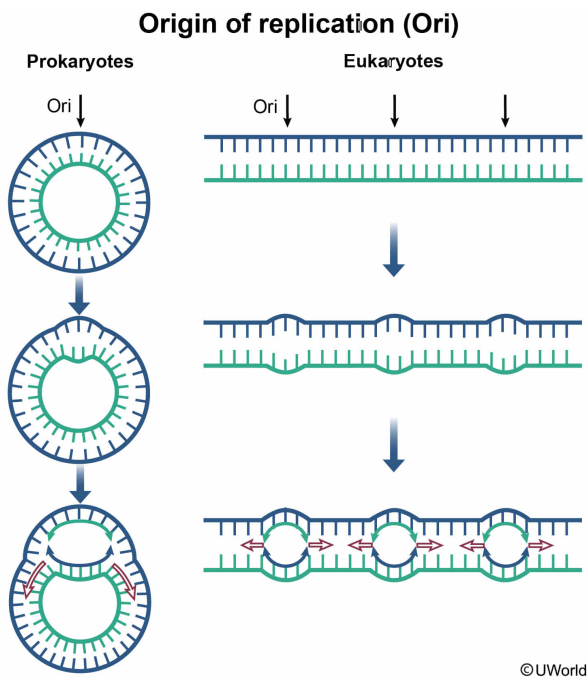
Which statement most likely explains the role of these peptides during replicative stress in the organisms studied?

- ☐ A. SGS1 would be required to repair replication forks in vivo at one origin of replication. [10%]
- ✓ ☒ B. recQ would be required to repair replication forks in vivo at one origin of replication. [47%]
- ☐ C. BLM would be unable to repair replication forks in vivo at multiple origins of replication. [21%]
- ☐ D. RECQL would be unable to repair replication forks in vivo at multiple origins of replication. [20%]

Correct

46% answered correctly

Explanation:



Homologous regions are areas of **protein/gene sequence similarity** between organisms from the same or different species.

In this question, researchers compared the similarity of the BLM peptide (ie, protein) sequence to its homologous counterparts in yeast, bacteria, and humans. A **protein's** amino acid sequence determines the function of that protein. Therefore, the finding that amino acid identity similarity (**42%-43%**) concentrated only to the seven DNA helicase domains of BLM, SGS1, recQ, and RECQL indicates that DNA helicase activity and function are likely evolutionarily conserved across these species.

Paragraph 1 in the passage states that BLM may be involved in DNA synthesis repair of stalled replication forks during replication stress (arrest). Although all the peptides obtained from the database can likely repair stalled replication forks, the recQ found in *Escherichia coli* (ie, the bacterial species) is the only option that would be required to repair stalled replication forks at only *one origin of replication* in vivo. This is because

prokaryotes have **circular DNA** with a **single origin of replication (Ori)** in the cytoplasm, whereas **eukaryotes** have **linear DNA** with **multiple origins of replication** in the nucleus. Therefore, a single *E. coli* bacterium would have circular DNA and a single origin of replication containing two replication forks, which would be repaired by recQ in the setting of replication stress.

(Choice A) The yeast (eukaryotic) SGS1 would likely repair stalled replication forks at *multiple* origins of replication in vivo.

(Choices C and D) BLM and RECQL are human (eukaryotic) genes with conserved DNA helicase activity that *would likely be found to repair* replication forks at the *multiple* replication origins present in a eukaryotic organism. The eukaryotic genome is significantly larger and more complex than the prokaryotic genome. Therefore, simultaneous replication at multiple origins allows the large eukaryotic genome to be copied in a rapid and effective manner despite its size.

Educational objective:

Prokaryotes typically have circular DNA with a single origin of replication in the cytoplasm whereas eukaryotes have linear DNA with multiple origins of replication in the nucleus. An origin of replication expands to form a replication bubble, which contains two replication forks that move apart in opposite directions during DNA synthesis.

Time spent:0 sec

Subject:
Biology

QID:400245

System:
Molecular Biology

The Bloom syndrome helicase (BLM) is catalytically inactive in Bloom syndrome (BS), a genetic disorder that leads to genomic instability characterized by excessive genetic recombination events in somatic cells. Research studies have indicated that BLM is a protein known as a replication stress response factor that aids in DNA synthesis repair of stalled replication forks during replication stress (arrest). To investigate this further, an experiment was conducted using cell lines derived from a patient with BS. Two groups of cells were generated, which initially differed only by BLM status. BLM⁺ cells were generated using cDNA cloning (ie, vector with BLM cDNA transfected). In contrast, BLM⁻ cells were transfected with an empty vector. Specific mutations, BLM-K695T and BLM-T99A, were also studied in stable transfected cells.

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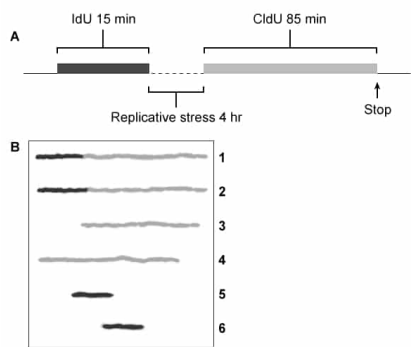


Figure 1 (A) Experimental protocol and (B) fluorescent imaging of a microscope slide showing incorporation of IdU (black lines) and CldU (gray lines) into DNA fibers isolated from a group of BLM⁺ and BLM⁻ cells, with single fibers depicted in each row

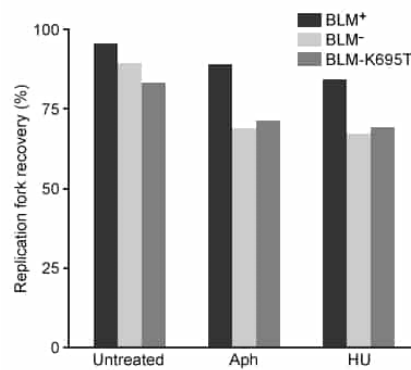


Figure 2 Recovery of stalled replication forks in cells expressing BLM⁺, BLM⁻, and the mutation BLM-K695T (antagonistic to wild-type)

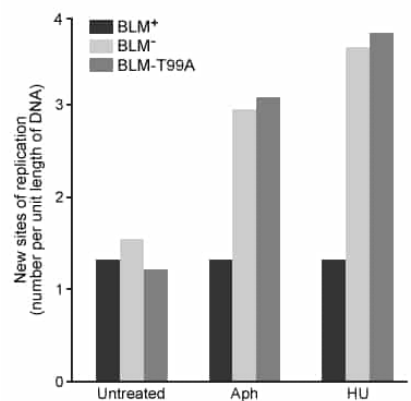
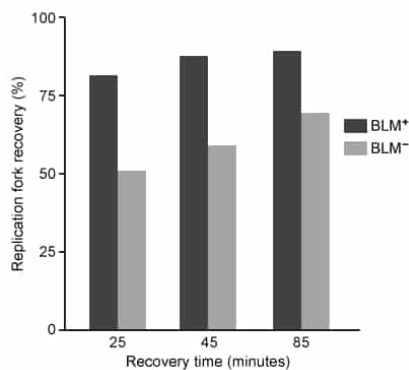


Figure 3 New sites of replication in cells expressing BLM⁺, BLM⁻, and the mutation BLM-T99A (T99 phosphorylated after

replication arrest)

After the 4-hour Aph treatment, recovery of stalled replication forks in BLM⁺ and BLM⁻ cells was analyzed as a function of time. Which of the following conclusions best explains the data shown in the graph below?

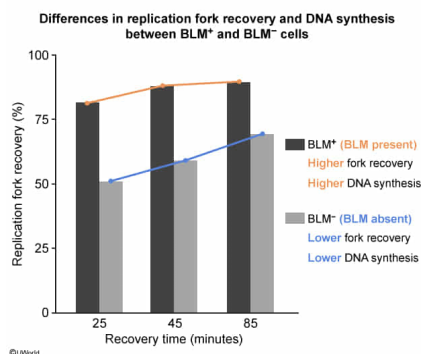


- ☐ A. The time at which replication fork recovery was assessed serves to predict BLM expression levels in both cell types. [17%]
- ✓ ☒ B. The rates of DNA synthesis at recovering replication forks differs in the two cell types. [54%]
- ☐ C. BLM⁻ cells are less sensitive to the effects of Aph than BLM⁺ cells. [16%]
- ☐ D. BLM activity in the two groups is primarily dependent on the percentage of replication forks that can be rescued. [11%]

Correct

54% answered correctly

Explanation:



In the experiment, researchers inhibited replication for 4 hours by treating BLM⁺ (BLM present) cells and BLM⁻ (BLM absent) cells with Aph and then analyzed the recovery of replication forks for 85 minutes. The graph shows that over time, BLM⁻ cells continued to have a **lower replication fork recovery rate** (ie, lower percentage of replication fork recovery) than BLM⁺ cells.

This difference in replication fork recovery between BLM⁺ and BLM⁻ indicates that rates of **DNA synthesis** at recovering forks likely also **differed** between the two cell types.

(Choice A) The passage indicates that BLM is *already absent* in BLM⁻ cells; therefore, the time when replication fork recovery is assessed does *not* provide any relevant information about BLM expression. In addition, the time points assessed correspond only to replication fork recovery rates and therefore cannot conclusively predict BLM *expression* levels.

(Choice C) Compared to BLM⁺ cells that express active BLM, BLM⁻ cells are *more sensitive* (ie, more damaged/affected) to Aph treatment because they lack the BLM transcript necessary for adequate replication fork recovery in the setting of replication stress.

(Choice D) BLM activity (ie, BLM presence or absence) is the **independent variable**, which was purposely altered by researchers in the experiment to assess its effect on the dependent variable (ie, replication fork recovery percentage over time). Therefore, when considering the *two groups of cells*, the *percentage of replication forks* that can be rescued *depends primarily on the presence or absence of BLM*.

Educational objective:

DNA synthesis can be arrested by replication inhibitors. Cells lacking the genes necessary to promote replication stability are more sensitive (damaged or affected) to the replication blockade induced by these agents, and recovery of replication may be impaired.

Time spent:0 sec

Subject:

Biology

QID:400246

System:

Molecular Biology

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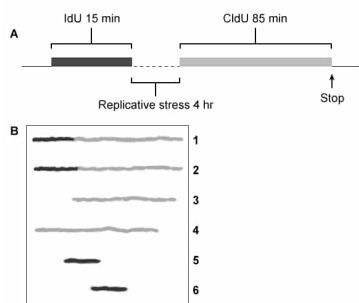


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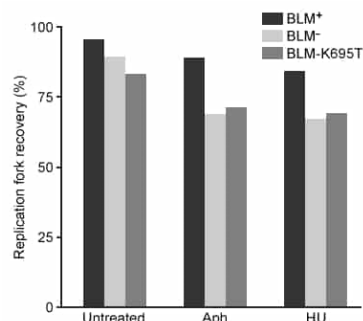


Figure 2 Recovery of stalled replication forks in cells expressing BLM⁺, BLM⁻, and the mutation BLM-K695T (antagonistic to wild-type)

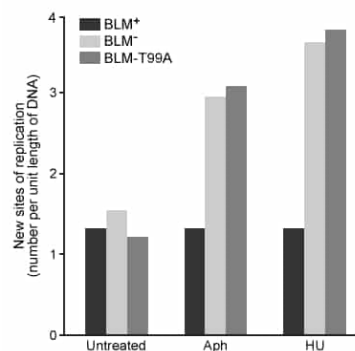


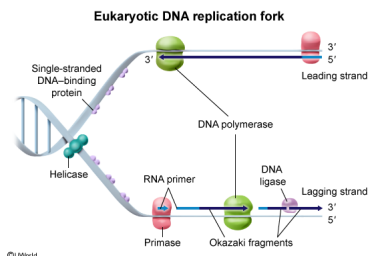
Figure 3 New sites of replication in cells expressing BLM⁺, BLM⁻, and the mutation BLM-T99A (T99 phosphorylated after replication arrest)

Based on the passage, what is the most accurate description of Aph-treated mutant cells compared to Aph-treated BLM⁺ cells?

- ☐ A. Supercoiling function of DNA topoisomerase is deficient in BLM-T99A cells during recovery of stalled replication forks. [13%]
- ☒ B. Binding of single-stranded DNA-binding proteins is increased in BLM-T99A cells at newly initiated replication forks. [57%]
- ☐ C. Unwinding activity by DNA helicase is increased in BLM-K695T cells during recovery of stalled replication forks. [22%]
- ☐ D. DNA primer production by primase is unchanged in BLM-K695T cells at newly initiated replication forks. [6%]

Correct
56% answered correctly

Explanation:



DNA replication within a **eukaryotic cell** is regulated by several key enzymes:

- **Topoisomerase** introduces negative supercoiling in the DNA double helix ahead of the replication fork to reduce the strain produced by unwinding, which causes positive supercoiling.
- **DNA helicase** unwinds the DNA double helix and separates the parent strands at the origin of replication.
- **Single-stranded DNA-binding protein** binds to each strand to prevent spontaneous reannealing of unwound single-stranded DNA.
- **Primase** synthesizes RNA primers and positions them at the beginning of each DNA strand. Only one primer is needed for leading strand synthesis, but lagging strand synthesis requires many RNA primers.
- **DNA polymerase** synthesizes daughter strands in a 5' → 3' direction only. One daughter strand is synthesized continuously toward the replication fork (leading strand); the other strand is synthesized discontinuously in a direction away from the replication fork (lagging strand), with more and more segments added as the replication fork progresses. This process results in the formation of Okazaki fragments, short stretches of newly synthesized DNA separated by RNA primers.
- **DNA ligase** joins Okazaki fragments.

In the passage, DNA replication was studied in mutant cells (ie BLM-T99A or BLM-K696T cells). The passage states that Aph was used to induce replication stress. Figure 3 shows that after Aph treatment, **BLM-T99A cells** have more new replication forks than BLM⁺ cells. Therefore, compared to BLM⁺ cells, **BLM-T99A cells** likely have **increased activity** of DNA replication enzymes, including **single-stranded binding proteins**, at newly initiated replication forks.

(Choice A) Increased numbers of *new* replication forks in Aph-treated BLM-T99A cells compared to Aph-treated BLM⁺ cells indicates that DNA topoisomerase function is likely *increased* in BLM-T99A cells at *newly initiated* forks. However, *no* data is provided regarding recovery of stalled replication forks in BLM-T99A cells.

(Choices C and D) In Figure 2, **BLM-K695T cells** have *decreased* replication fork recovery after Aph treatment compared to BLM⁺ cells. Therefore, replication enzyme (eg, DNA helicase, primase) activity in BLM-K695T cells is likely *decreased* during replication fork recovery. In addition, *no* data is provided regarding new replication fork initiation in BLM-K695T cells.

Educational objective:

DNA replication occurs in the 5' → 3' direction; DNA helicase first unwinds the double helix and separates the parent strands. Single-strand DNA-binding proteins prevent spontaneous reannealing of unwound single-stranded DNA. Leading strand synthesis is continuous, whereas synthesis of the lagging strand (composed of RNA primers and Okazaki fragments) is discontinuous.

Time spent: 0 sec
Subject:
Biology

QID: 400247
System:
Molecular Biology

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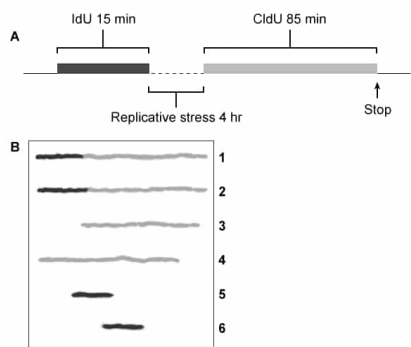


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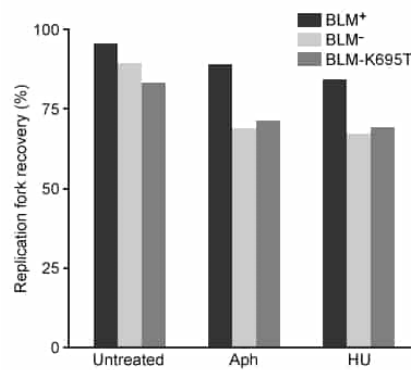


Figure 2 Recovery of stalled replication forks in cells expressing BLM⁺, BLM⁻, and the mutation BLM-K695T (antagonistic to wild-type)

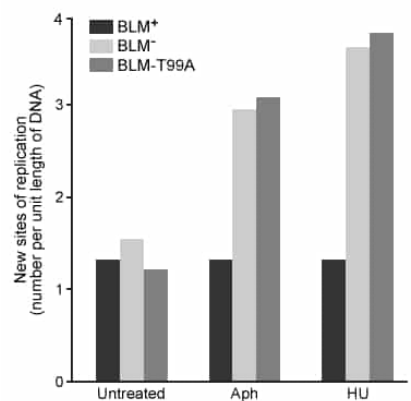


Figure 3 New sites of replication in cells expressing BLM⁺, BLM⁻, and the mutation BLM-T99A (T99 phosphorylated after
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replication arrest)

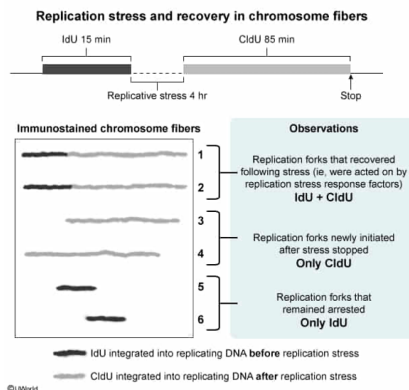
In Figure 1B, which rows most likely represent DNA fibers that were successfully acted upon by replication stress response factors during the experiment?

- ✓ ☒ A. Rows 1 and 2 [57%]
☐ B. Rows 2 and 5 [2%]
☐ C. Rows 3 and 4 [17%]
☐ D. Rows 5 and 6 [22%]

Correct

57% answered correctly

Explanation:



The passage describes DNA replication stress in which replication forks are stalled (ie, arrested). Accordingly, during this replication stress, DNA synthesis is arrested at the S phase of the **eukaryotic cell cycle**. In addition, the passage states that BLM is a replication stress response factor involved in repair of stalled replication forks.

In the passage experiment, cells (either BLM⁺ or BLM⁻) were initially treated with IdU (ie, a nucleoside analog), which could be incorporated into replicating DNA strands. Next, replication stress was induced for four hours with either hydroxyurea (HU) or aphidicolin (Aph). Following these four hours of stress, cells were treated with CldU, which could be incorporated into DNA strands replicating after stress occurred. Figure 1B shows DNA strands formed in cells during this experiment.

Because BLM⁺ cells would have normal BLM function, this replication stress response factor could act to recover stalled replication forks in these cells. **Rows 1 and 2** in Figure 1B show strands that, after the four hours of replication stress, recovered and continued replicating (ie, incorporated CldU), indicating that **replication stress response factors** (eg, BLM) were likely acting successfully at these strands.

(Choices B and D) Rows 5 and 6 show DNA strands in which DNA replication occurred initially (ie, initial IdU incorporation occurred). However, after the replication stress, DNA replication was never recovered (ie, CldU was never incorporated). Therefore, replication stress response factors likely did *not* act successfully on these strands to restart replication.

(Choice C) Rows 3 and 4 show only CldU incorporation, which means that these *new sites of replication* arose *after* replication stress had already stopped. Therefore, replication stress response factors likely did *not* need to act at these strands.

Educational objective:

DNA replication (synthesis) occurs in S phase of the eukaryotic cell cycle.

Time spent: 0 sec
Subject:
Biology

QID: 400248
System:
Molecular Biology

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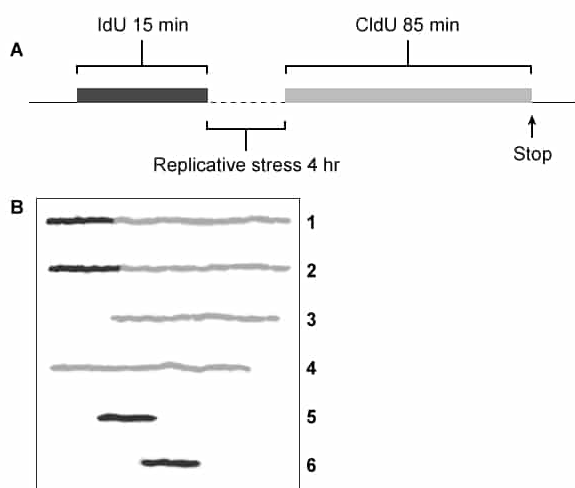


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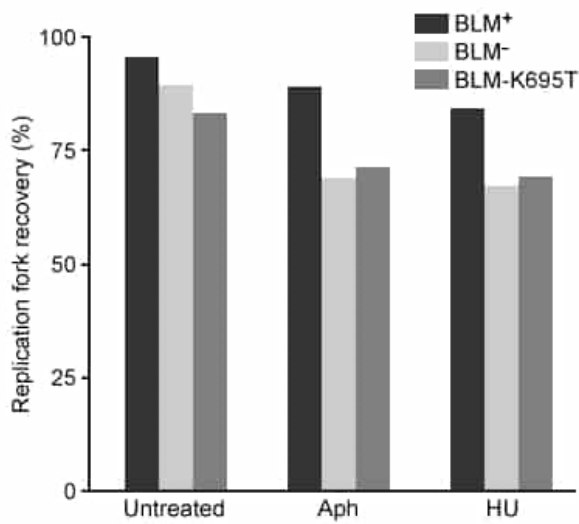


Figure 2 Recovery of stalled replication forks in cells expressing BLM⁺, BLM⁻, and the mutation BLM-K695T (antagonistic to wild-type)

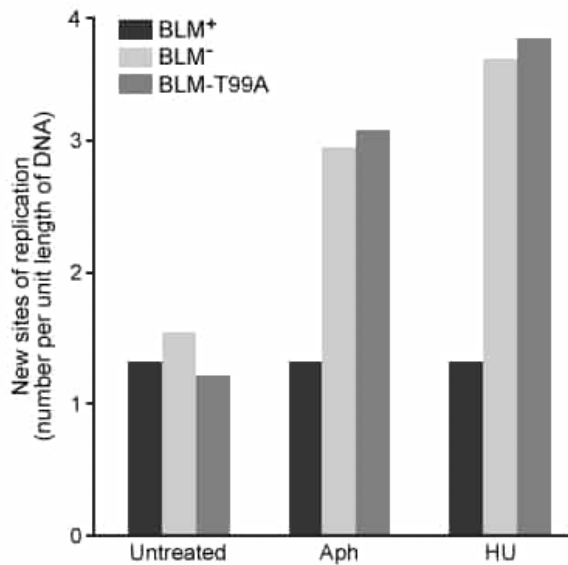
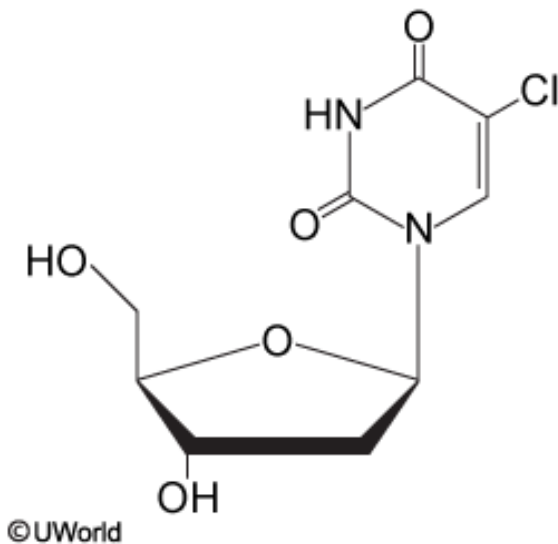


Figure 3 New sites of replication in cells expressing BLM⁺, BLM⁻, and the mutation BLM-T99A (T99 phosphorylated after replication arrest)

After the 85-minute period described in the passage, deoxyribonucleoside triphosphates (dNTPs) are added and DNA synthesis is allowed to continue for an additional 20 minutes. Which of the following processes is most likely to occur during the new replication period? The structure of CldU is shown below.

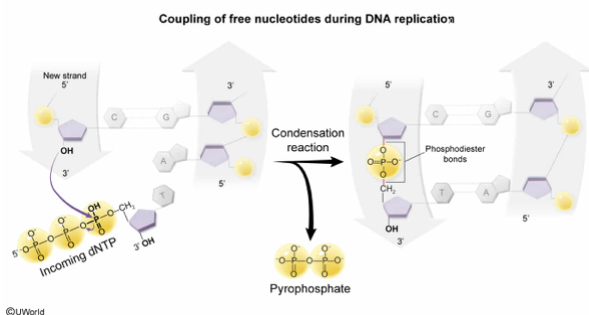


- ☐ A. CldU and a dNTP molecule would react endergonically to form a nucleic acid, releasing two free phosphate groups in the process. [9%]
- ☐ B. CldU and a dNTP molecule would react exergonically to form a covalent phosphodiester bond, consuming a water molecule in the process. [19%]
- ✓ ☒ C. The 3' OH group from the last nucleotide of the growing strand would attack the 5' PO₄ group of an incoming dNTP. [51%]
- ☐ D. The 5' PO₄ group from the last nucleotide of the growing strand would attack the 3' OH group of an incoming dNTP. [18%]

Correct

50% answered correctly

Explanation:



During replication, **DNA polymerase** attaches uncoupled **deoxyribonucleoside triphosphates (dNTPs)** to the growing DNA strand. Each dNTP (dTTP, dATP, dCTP, and dGTP) is composed of a base, a deoxyribose sugar, and three phosphate (PO₄) groups. The passage states that CldU is a **thymidine analog**, meaning that it can be incorporated into the growing DNA strand in place of thymidine, which bonds to

adenosine via two hydrogen bonds. After 85 minutes, the time-controlled incorporation of CldU into the growing DNA strand stops. Therefore, thymidine would be incorporated into the growing strand instead of CldU during the new 20-minute replication period.

Based on the AT/CG rule of [complementary base pairing](#), a free dNTP enters the catalytic site of DNA polymerase and forms hydrogen bonds with a complementary nucleotide on the parent (template) DNA strand. The **3' OH** (*not* 5' PO₄) **from the last nucleotide** of the new growing strand **"attacks" the 5' PO₄** (*not* 3'OH) **group** of the incoming dNTP (**Choice D**).

As a result, two phosphate groups are released from the dNTP, resulting in a nucleoside monophosphate molecule attached to the growing strand and in the release of **pyrophosphate (PP_i)** and energy (exergonic process). PP_i is further broken down into two phosphate groups to release more energy. The energy generated from these reactions is used to form a **covalent phosphodiester bond** between the last nucleotide of the growing strand and the incoming dNTP. Once the phosphodiester bond is formed, the enzyme proceeds to couple the next free dNTP to the new growing strand.

(Choice A) When a dNTP molecule (or analog such as CldU) is added to a growing DNA strand, a pyrophosphate molecule is released. This pyrophosphate is then hydrolyzed in an exergonic (*not* endergonic) manner, releasing two free phosphate groups and energy.

(Choice B) Water is consumed in [hydrolysis reactions](#), not condensation reactions such as DNA synthesis.

Educational objective:

During replication, DNA polymerase joins uncoupled deoxyribonucleoside triphosphates to the new DNA strand. As a result, the exergonic release of a pyrophosphate molecule leads to the formation of a covalent phosphodiester bond between the last nucleotide of the growing strand and the incoming nucleotide.

Time spent:0 sec

Subject:
Biology

QID:400249

System:
Molecular Biology

The Bloom syndrome helicase (BLM) is catalytically inactive in Bloom syndrome (BS), a genetic disorder that leads to genomic instability characterized by excessive genetic recombination events in somatic cells. Research studies have indicated that BLM is a protein known as a replication stress response factor that aids in DNA synthesis repair of stalled replication forks during replication stress (arrest). To investigate this further, an experiment was conducted using cell lines derived from a patient with BS. Two groups of cells were generated, which initially differed only by BLM status. BLM⁺ cells were generated using cDNA cloning (ie, vector with BLM cDNA transfected). In contrast, BLM⁻ cells were transfected with an empty vector. Specific mutations, BLM-K695T and BLM-T99A, were also studied in stable transfected cells.

The nucleoside (thymidine) analogs iododeoxyuridine (IdU) and chlorodeoxyuridine (CldU) were used to label and track genomic regions of active replication. Cells were initially treated with IdU for 15 minutes. Subsequently, either 4 mM of hydroxyurea (HU) or 30 μ M of aphidicolin (Aph) were used to induce replication stress for 4 hours, after which cells were treated with CldU for another 85 minutes to assess replication recovery. Cells were mounted on microscope slides, lysed to isolate chromosome fibers, and then fixed on the slides. Sites of IdU and CldU incorporation into the growing DNA strand were then detected by immunostaining the isolated chromosome fibers with analog-specific, fluorescent-coupled antibodies. Replication activity was visualized via microscopy and measured.

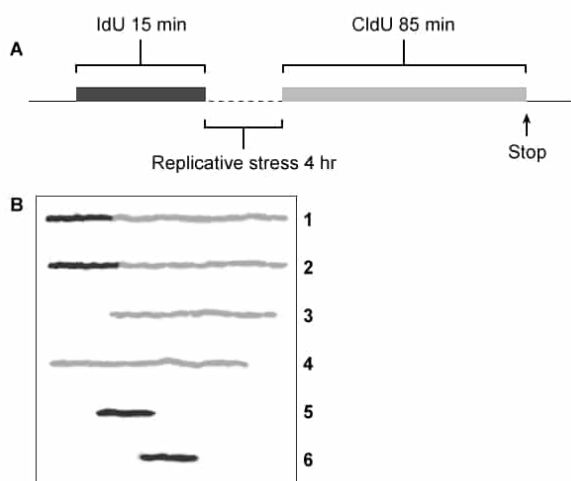


Figure 1 (A) Experimental protocol and (B) fluorescent imaging of a microscope slide showing incorporation of IdU (black lines) and CldU (gray lines) into DNA fibers isolated from a group of BLM⁺ and BLM⁻ cells, with single fibers depicted in each row

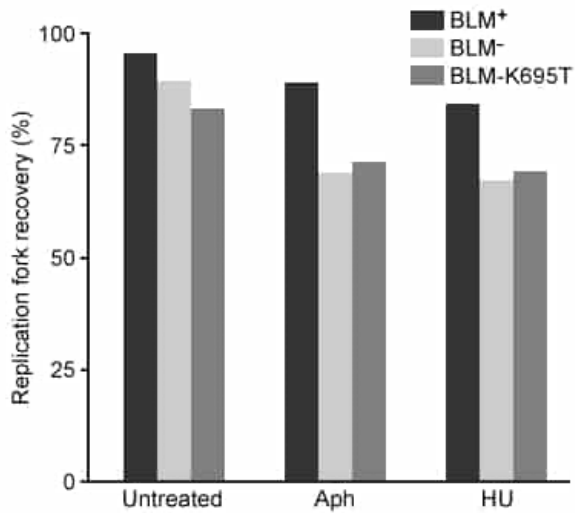


Figure 2 Recovery of stalled replication forks in cells expressing BLM⁺, BLM⁻, and the mutation BLM-K695T (antagonistic to wild-type)

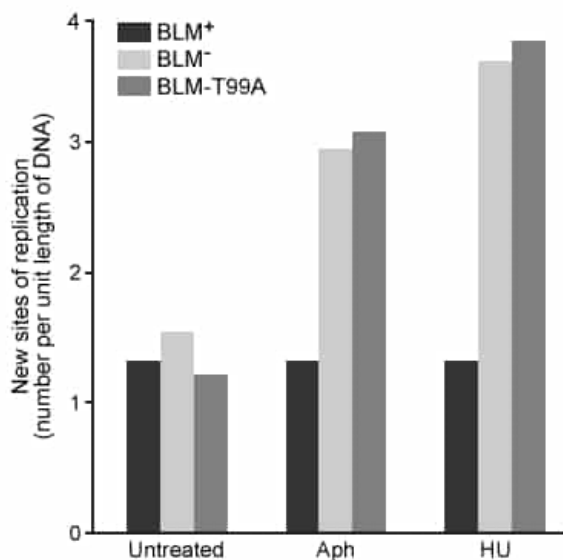


Figure 3 New sites of replication in cells expressing BLM⁺, BLM⁻, and the mutation BLM-T99A (T99 phosphorylated after replication arrest)

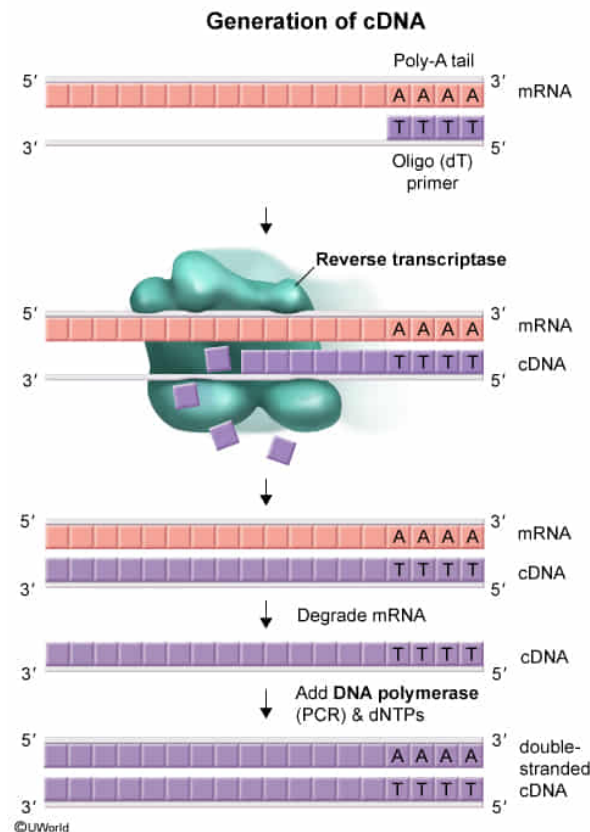
All of the following enzymes would be directly used to generate the vector transfected into BLM⁺ cells EXCEPT:

- ☐ A. Reverse transcriptase [18%]
- ☒ B. RNA polymerase [44%]
- ☐ C. DNA polymerase [18%]
- ☐ D. Restriction enzyme [19%]

Correct

44% answered correctly

Explanation:



During **cDNA cloning**, a messenger RNA (mRNA) transcript of interest is identified in an organism. The mRNAs can then be isolated and mixed with complementary primers known as oligo(dT) primers, which are primers composed of thymine nucleotides. These primers bind the mRNA 3' **poly-A tails**, and the enzyme **reverse transcriptase** uses **dNTPs** to generate single-stranded cDNA molecules from target mRNAs (**Choice A**). The mRNA strand is then degraded and the resulting cDNA (eg, *BLM* cDNA) is amplified using **DNA polymerase** and **PCR** (**Choice C**).

The cDNA sequence containing the gene of interest can then be inserted into a **cloning vector**, a circular DNA molecule found in bacteria and viruses. The foreign cDNA and vector are cut by a **restriction enzyme** (**Choice D**), generating complementary sticky ends that anneal when both molecules are mixed together. **DNA ligase** joins the cDNA to the vector, resulting in the formation of a recombinant vector that can be transfected (inserted) into cells, as was done with the *BLM* cDNA in *BLM*⁺ cells in the passage.

RNA polymerases synthesize various types of RNAs but are not used as standard reagents in cDNA cloning because the target mRNA is already present in the cell (ie, does *not* need to be synthesized).

Educational objective:

In cDNA cloning, reverse transcriptase generates a single strand of cDNA from a target mRNA sequence. DNA polymerase synthesizes the second complementary DNA strand and amplifies the target cDNA sequence. The target cDNA can then be inserted into a plasmid vector via the actions of a restriction enzyme (cuts both plasmid and

vector) and DNA ligase (joins the cDNA to the vector).

Time spent:0 sec

Subject:

Biology

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